GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-001 Distribution Safety Rules - Summary for Operational Staff

Field	Value
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Version	6
Effective from	2025-01-06
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Owner	Electricity Safety Manager
Approved by	Head of Electricity Network Operations
Applies to	All GigaWhat Electricity sites, plant and equipment

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This document summarises the GigaWhat Electricity Distribution Safety Rules for staff who operate, inspect or work on the electricity network. It is a quick reference. The detailed requirements are in the procedures listed in section 3, and where this summary and a detailed procedure differ, the detailed procedure applies.

2 Scope

This summary applies to all employees and contractors who hold an authorisation under the Distribution Safety Rules, at every GigaWhat Electricity site: Riverside Lane Grid Substation (ESS-RVL), Ashford Road Primary Substation (ESS-ASH), Kingsmead Primary Substation (ESS-KGM) including the Kingsmead cable tunnel (CT-KGM-01), Brookfield Secondary Substation (ESS-BRK), Orchard Way Secondary Substation (ESS-ORC) and the Ashford Road depot (DEP-ASH). It covers high voltage (132 kV, 33 kV and 11 kV) and low voltage equipment.

3 References

- PR-ELEC-012 Isolation and Earthing of 11 kV Circuits
- PR-ELEC-015 Switching Programmes on the 11 kV and 33 kV Networks
- PR-ELEC-020 Permit-to-Work and Sanction-for-Test (Electrical)
- PR-ELEC-031 Entry to Kingsmead Cable Tunnel (Confined Space)
- PR-ELEC-063 Substation Access, Keys and Visitors
- PR-CORP-001 Reporting Incidents and Near Misses

- PR-CORP-004 Lone Working
- HSE HSG85 Electricity at work: Safe working practices
- HSE HSR25 Memorandum of guidance on the Electricity at Work Regulations 1989
- Electricity at Work Regulations 1989
- Electricity Safety, Quality and Continuity Regulations 2002
- Management of Health and Safety at Work Regulations 1999

4 Definitions

High voltage (HV): a voltage above 1000 V AC. On the GigaWhat Energy network this means 132 kV, 33 kV and 11 kV.

Low voltage (LV): a voltage of 1000 V AC or below.

Dead: at or about zero voltage and disconnected from any live system.

Isolated: disconnected from every source of supply by an open switching device that is secured against reconnection.

Earthed: connected to the general mass of earth so that any stored or induced energy is discharged immediately.

Safety document: a Permit-to-Work or Sanction-for-Test issued under PR-ELEC-020.

Authorisation: a written appointment by the Electricity Safety Manager stating the duties a person may carry out under the Distribution Safety Rules.

5 Responsibilities

Head of Electricity Network Operations: approves the Distribution Safety Rules and provides the resources needed for training and authorisation.

Electricity Safety Manager: maintains the rules and the Authorisation Register, and approves voltage indicators, earthing equipment and personal protective equipment.

Senior Authorised Person (HV): approves complex isolations and 33 kV switching programmes, and audits safety documents.

Authorised Person (HV): carries out switching, isolation and earthing, and issues and cancels safety documents.

Competent Person (HV): receives safety documents, carries out work under them and may carry out routine inspections.

Control Engineer: controls the HV network, approves 11 kV switching programmes, instructs each switching operation and issues safety document numbers.

All staff: follow these rules, stop work when it is unsafe and report incidents and near misses.

6 Procedure

6.1 Authorisation and competence

6.1.1 Carry out only the duties shown on your current written authorisation, as recorded in the Authorisation Register held by the Electricity Safety Manager.

6.1.2 Carry your authorisation card on site and show it when asked by an Authorised Person (HV), Senior Authorised Person (HV) or Control Engineer.

6.1.3 Complete refresher training and reassessment at least every three years. An authorisation that is not renewed lapses automatically.

6.1.4 If you are unsure whether a task is within your authorisation, stop and ask an Authorised Person (HV) or Senior Authorised Person (HV) before starting.

6.1.5 Do not instruct or allow anyone to carry out duties beyond their authorisation.

6.1.6 Tell your manager and the Electricity Safety Manager about any medical condition or medication that could affect your fitness to carry out authorised duties.

6.2 Basic rules for HV equipment

> WARNING: Touching, or approaching too close to, a live HV conductor can cause a fatal electric shock or severe arc flash burns. HV equipment must be treated as live until it has been made safe under a safety document.

6.2.1 Treat all HV equipment as live until it has been isolated, proved dead, earthed and a safety document issued for it.

6.2.2 Do not work on or near live HV conductors. Live working on HV equipment is prohibited on the GigaWhat Energy network.

6.2.3 Keep at least the safety distance from exposed live HV conductors: 0.8 m at 11 kV, 1.0 m at 33 kV and 1.4 m at 132 kV.

6.2.4 Do not defeat, override or bypass any interlock, shutter or locking device.

6.2.5 Wear arc-rated clothing rated at not less than 12 cal/cm², a face visor and insulating gloves when operating HV switchgear.

6.2.6 Use only voltage indicators, earthing equipment, test instruments and tools of a type approved by the Electricity Safety Manager.

6.3 Switching and operation

6.3.1 Carry out HV switching only under the control of the Control Engineer and in accordance with a switching programme prepared under PR-ELEC-015.

6.3.2 Before each operation, read the equipment label and check it against the switching programme.

6.3.3 Repeat back each instruction from the Control Engineer and wait for confirmation before acting on it.

> WARNING: The oil circuit breakers on SWG-KGM-11 at ESS-KGM are under an operational restriction. A failure during operation could cause an oil fire or explosion and kill anyone standing in front of the panel.

6.3.4 Operate SWG-KGM-11 circuit breakers only remotely from the control room, or with the portable remote operating device from outside the switchroom, with nobody inside.

6.3.5 Operate HV switchgear without the Control Engineer's instruction only in an emergency to prevent danger to life, and report the operation to the Control Engineer immediately afterwards.

6.4 Safety documents

6.4.1 Isolate and earth HV equipment in accordance with PR-ELEC-012 before any safety document is issued.

6.4.2 Issue Permits-to-Work and Sanctions-for-Test only in accordance with PR-ELEC-020. Only an Authorised Person (HV) may issue them.

> HOLD POINT: Work must not start until the Authorised Person (HV) has issued the safety document and shown the Competent Person (HV) the safety precautions and the point of work, and the Competent Person (HV) has signed the receipt.

6.4.3 As the Competent Person (HV), brief every member of the working party on the safety document and make sure each signs on before work starts.

6.4.4 Work only within the limits of the safety document. If the work needs to change, stop and contact the issuing Authorised Person (HV).

6.4.5 Never remove a Circuit Main Earth, safety lock or notice unless the safety document allows it.

6.5 Access, confined spaces and lone working

6.5.1 Enter substations only in accordance with PR-ELEC-063, signing the site log and informing the Control Engineer on arrival and departure.

6.5.2 Enter the Kingsmead cable tunnel CT-KGM-01 only under a confined space entry permit in accordance with PR-ELEC-031.

6.5.3 Work alone only for routine inspection and for switching agreed with the Control Engineer, following PR-CORP-004. Never carry out work under a Permit-to-Work alone.

6.5.4 Keep substation doors and gates locked when unattended, and never leave a site open to the public.

6.6 Stopping work and reporting

6.6.1 Stop work and make the area safe if conditions change, if equipment behaves unexpectedly or if you are in any doubt about safety.

6.6.2 Report every incident, near miss and dangerous occurrence under PR-CORP-001 before the end of your shift.

6.6.3 Report damaged or defective equipment to the Control Engineer straight away, and withdraw defective safety equipment from use.

6.7 Emergency

6.7.1 Call 999 for any injury or fire, then inform the Control Engineer.

6.7.2 If someone has received an electric shock, do not touch them until the supply has been isolated or they have been moved clear by insulated means. Start first aid and use a defibrillator if one is available.

6.7.3 Do not fight a fire in electrical equipment unless it is safe to use a carbon dioxide extinguisher. Leave the building and wait for the fire and rescue service at the site entrance.

6.7.4 Ask members of the public who report a power cut or damaged power lines to call 105, or 999 if there is immediate danger.

7 Records

The Electricity Safety Manager keeps the Authorisation Register and training records for the life of each authorisation plus six years. Safety documents and switching programmes are kept as set out in PR-ELEC-020 and PR-ELEC-015. Incident reports are kept under PR-CORP-001.

8 Revision history

Version	Date	Change
1	2010-04-06	First issue.
2	2013-03-04	Added summary of Sanction-for-Test arrangements.
3	2016-01-11	Linked authorisations to the new Authorisation Register. Added confined space entry requirements.
4	2019-01-07	Added arc-rated clothing for HV switching.
5	2022-01-10	Added remote operation of SWG-KGM-11 oil circuit breakers. Aligned lone working limits with PR-CORP-004.
6	2025-01-06	Three-yearly review. Added reference to PR-ELEC-031 for the Kingsmead cable tunnel. Stated that live working on HV equipment is prohibited. Updated safety distances and added the 105 power cut number to the emergency section.